

Unifying spatially-explicit Rao's entropy, spatial concentration, and accumulation curves: the different aspects of local biodiversity

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Abstract

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1. Introduction

Local biotic and abiotic interactions are for long recognized as fundamental processes that shape the structure of ecological communities. In particular, the spatial pattern of plant communities may result from the combined effects of seed dispersal, inter-individual (i.e. intra- and inter-specific) competition/facilitation and habitat preferences (Dale 1999). While each single process may either promote or demote species coexistence, the diversity of interactions in space and/or time is a key feature for the maintenance of biological diversity (Clark et al. 2010; Cadotte et al. 2012). As a consequence, the ecological literature reveals increasing efforts in integrating approaches from phylogenetics, functional and community ecology (Mouquet et al. 2012), in order to account for the key components of species interactions with their

biotic and abiotic environments.

Towards this goal, a number of novel analytical methods incorporating measures of between-species phylogenetic relatedness or functional trait dissimilarities as a surrogate for ecological differences have been developed (Pavoine and Bonsall 2011). Among these methods some are specifically designed to explicitly account for space (e.g. Pavoine et al. 2011). However those studies generally consider spatially discontinuous community samples, i.e. the β -diversity component of Whittaker (1960), so that the underlying spatial framework can basically be related to the graph theory (Dale et al. 2002). When mapped data are concerned, such as for instance those on tropical trees provided by large permanent forest plots (Losos and Leigh 2004), the well-developed framework of the spatial point pattern analysis (Ripley 1976, 1977) offers an efficient means of studying the continuous spatial variation in species diversity (Condit et al. 2000; Shimatani 2001a; Rajala and Illian 2012). A synthesis of the opportunities brought by this approach can be found in Wiegand et al. (2017).

Shen et al. (2013) and Pélissier and Goreaud (2015) concomitantly introduced within the framework of the spatial point pattern analysis a multitype correlation function that provides a distance-dependent measure of species diversity accounting for non-uniform species differences. If both contributions independently arrived to very similar

functions, we believe that to help ecologists to well-understand and to take full advantage of novel approach, further clarifications are needed, especially regarding how it is related to well-known functions of spatial point pattern analysis, namely Ripley's K-function (Ripley 1977) and its inter-type extension, the $K_{1,2}$ -function (Lotwick and Silverman 1982) on the one hand, and Rao's quadratic entropy on the other hand (Rao 1982).

The aim of this chapter is thus to introduce the rationale behind the functions of multivariate point pattern analysis that integrate between-species differences in a distance-dependent measure of diversity. It can basically be seen as an extension of the work of Shimatani (2001a) based on Simpson's entropy to Rao's entropy, one of the most powerful and flexible measure of species diversity (Pavoine 2012). It is also a particular implementation of the measurement of neighborhood diversity which possesses some specific features due to the quadratic form of Rao's entropy. The methods section hereafter makes the synthesis between these originally independent developments and outlines the central role of Rao's entropy to reconcile them. We finally provide worked examples based on simulated data and on a real dataset from an experimental tropical forest plot in French Guiana.

2. Methods

2.1 Biodiversity measurement based on information theory

2.2 Diversity accumulation curves

2.3 Ripley's K

The framework of spatial point pattern analysis, as introduced by Ripley (1977), relies on the K-function, defined so that $\lambda K(r)$ is the expected number of neighbours located within a distance r of an arbitrary point of the pattern with intensity λ . Its classical unbiased estimator for a point pattern located within a finite sampling window is

$$\hat{K}(r) = \frac{1}{\hat{\lambda}n} \sum_{i=1}^n \sum_{j=1}^n \mathbf{1}(\|x_i - x_j\| \leq r) c(i, j, r) \quad (1)$$

where n is the number of points of the pattern, $\hat{\lambda}$ is the pattern intensity generally estimated as n/A , with A the sampling window area. The indicator function $\mathbf{1}(\|x_i - x_j\| \leq r)$ equals 1 when points i and j are less than r apart, 0 else. $c(i, j, r)$ is

correction of edge effects equal to 1 when the circle of radius r fits into the sampling window. When it does not, the isotropic correction proposed by Ripley (1977) applies to windows of any shape thanks to appropriate geometrical considerations (Goreaud and Pélissier 1999) to estimate the number of neighbors with no bias.

A common extension of this univariate function is the intertype (or bivariate) K_{12} -function proposed by Lotwick and Silverman (1982). Its estimator is:

$$\hat{K}_{1,2}(r) = \frac{1}{\hat{\lambda}_2 n_1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \mathbf{1}(\|x_i - x_j\| \leq r) c(i, j, r) \quad (2)$$

$K(r)$ is a multiscale function of second-order neighbourhood properties of a point pattern, which allows robust tests of statistical significance against the null hypothesis of a random distribution of the points within the sampling window. In contrast, $K_{12}(r)$ allows characterizing the spatial interaction between the two types of points, which can be tested against various null hypotheses. The most interesting ones are the random labelling hypothesis, which means that, conditionally to the point locations, types 1 and 2 are randomly distributed among points of the pattern, and the population independence hypothesis, meaning that, conditionally to the species-specific spatial patterns, population of type 1 points is independent from population of type 2 points (Goreaud and Pélissier 2003).

2.4 A distance-dependent Simpson's entropy

Simpson's index of species diversity (Simpson 1949) gives the probability that a randomly selected pair of individuals in a sample is comprised of two different species. It is more than an index: it is a measure of entropy (Jost 2006) and thus should be named Simpson's entropy. It is estimated with no bias by:

$$\hat{D} = 1 - \sum_{s=1}^S \frac{n_s(n_s - 1)}{n(n - 1)} \quad (3)$$

where n and S are the total numbers of individuals and species in the sample, respectively, and n_s the number of individuals of species s .

From this entropy, Shimatani (2001a) proposed a distance-dependent version based on Ripley's

K-function, from which we can write a version standardized by $\hat{D}$ as

$$\hat{K}_S(r) = \left[1 - \sum_{s=1}^S \frac{\hat{\lambda}_s^2 \hat{K}_s(r)}{\hat{\lambda}^2 \hat{K}(r)} \right] / \hat{D} \quad (4)$$

where $\hat{K}(r)$ and $\hat{K}_s(r)$ are Ripley's K-functions for all points of the pattern (no distinction of species) and for points of species s , respectively, with $\hat{\lambda}$ and $\hat{\lambda}_s$ the corresponding intensity estimators.

The numerator of the above equation corresponds to function $\hat{\alpha}(r)$ of Shimatani (2001a), that we may name Simpson's neighbourhood entropy. Neighbourhood entropy (or diversity) is the entropy (or diversity) considered within a circle of radius r around each point. The interest of considering local diversity rather than at the scale of the whole window has emerged in the literature since Batty (1974). A review can be found in Altieri et al. (2021), with diverse approaches. It is not discussed that the functioning of a community is influenced by its biodiversity, but it is straightforward to consider that each individual is concerned by the diversity of its neighbourhood rather than of the whole community (e.g., Hubbell et al. 2001 showed that tree mortality increased with neighbourhood species richness in a tropical evergreen forest).

$\alpha(r)$ tends to D when r increases so $K(r) \rightarrow 1$. It is noticeable that $\hat{K}_S(r)$ has an expectation of 1 for all r under the random labelling hypothesis, i.e. a random distribution of the species names conditionally to the overall spatial pattern (Eckel et al. 2008).

Using the unbiased estimator of $\hat{K}(r)$ introduced in Equation 1, allows correcting for edge effects for sampling windows of any shape.

We can notice for the following that $\hat{K}_S(r)$ can also be estimated by considering the pairs of heterospecifics (Shimatani 2001a, eq. 9):

$$\hat{K}_S(r) = \left[\sum_{s=1}^S \sum_{t=1, s \neq t}^S \frac{\hat{\lambda}_s \hat{\lambda}_t \hat{K}_{s,t}(r)}{\hat{\lambda}^2 \hat{K}(r)} \right] / \hat{D} \quad (5)$$

with $\hat{K}_{s,t}(r)$ a generalization of $K_{12}(r)$ estimator introduced earlier, to any couple of different species s and t .

2.5 A generalization to Rao's entropy

Simpson's entropy is a special case of Rao's entropy, that is the expected divergence between individuals of a randomly selected pair in a sample (Rao 1982). An unbiased estimator (Shimatani 2001b) is:

$$\hat{H}_D = \sum_{s=1}^S \frac{n_s n_t}{n(n-1)} d_{s,t}$$

where and $d_{s,t}$ is a measure of the pairwise divergence between species s and t (with $d_{s,t} = d_{t,s}$ and $d_{s,s} = 0$, but the triangular inequality $d_{s,t} + d_{s,u} \geq d_{s,u}$ that defines a distance is not required). H_D is a measure of diversity commonly used in ecology to take into account non-uniform distances between species, such as phylogenetic or functional distances (Pavoine and Bonsall 2011). Taking $d_{s,t} = 1$ for all $s \neq t$ and 0 otherwise, H_D reduces to Simpson's D above. We therefore straightforwardly propose to generalize $K_S(r)$ introduced above to a normalized, distance-dependent Rao's quadratic entropy $K_S(r)$ that integrates Ripley's correction of edge effects estimated by:

$$\hat{K}_R(r) = \left[\sum_{s=1}^S \sum_{t=1, s \neq t}^S \frac{\hat{\lambda}_s \hat{\lambda}_t \hat{K}_{s,t}(r) d_{s,t}}{\hat{\lambda}^2 \hat{K}(r)} \right] / \hat{H}_D$$

This function has an expectation of 1 under the random labelling hypothesis and gives the mean proportion of Rao's sample diversity (i.e., H_D) in a disc of radius r centred on an arbitrary point of the pattern. $\hat{K}_R(r)$ reduces to $\hat{K}_S(r)$ when $d_{s,t} = 1$ for all species $s \neq t$.

2.6 Multitype correlation and β diversity

Our K_R - and K_S -functions are cumulative functions as is the original K-function of Ripley (1977). This means that they are estimated for a series of concentric neighbourhood discs of increasing radii, for instance by a regular distance interval dr . Some authors however prefer to refer to the framework of the pair-correlation function, $g(r)$ (Stoyan et al. 1987; Wiegand and Moloney 2004), that can be seen as the derivative of $K(r)$, giving the probably to encounter neighbouring points in concentric annuli between r and $r + dr$. $K_d(r)$ of Shen et al. (2013) is such a function, very close to $g_R(r)$, the derivative of our $K_R(r)$ above (Pélissier and Goreaud 2015). $K_d(r)$ is therefore an

awkward appellation for Shen's et al. function, which would have been better named as $g_d(r)$ as to remain consistent with the classical notation used in point pattern analysis.

The same reasoning can be built from the so-called Simpson's β diversity index to $g_R(r)$. Chave and Leigh (2002) defined the $F(r)$ function as the probability for two individuals located r apart to belong to the same species. Shen et al. (2013) named $1 - F(r)$ the 'spatially explicit Simpson's index of diversity', denoted $\beta(r)$ by Shimatani (2001a), that is the probability for two individuals located r apart to belong different species. We will rather write $\beta_S(r)$ for consistency with our notation system. It can be straightforwardly be generalized to $\beta_R(r)$, the average divergence between to individuals taken at distance r , thus a spatially explicit Rao's index of diversity. These measures are actually β differentiation diversity measures sensu Jurasinski et al. (2009): they characterize how different spatial units r apart are.

We can thus define $g_S(r)$, a normalized version of $\beta_S(r)$ and $g_R(r)$, its generalization to Rao's entropy.

3. Application

3.1 Simulated data

3.2 Paracou

4. Conclusion

References

- Altieri, Linda, Daniela Cocchi, and Giulia Roli. 2021. 'Spatial Entropy for Biodiversity and Environmental Data: The R-package SpatEntropy'. *Environmental Modelling & Software* 144 (October): 105149. <https://doi.org/10.1016/j.envsoft.2021.105149>.
- Batty, Michael. 1974. 'Spatial Entropy'. *Geographical Analysis* 6 (1): 1–31. <https://doi.org/10.1111/j.1538-4632.1974.tb01014.x>.
- Cadotte, Marc W., Russel Dinnage, and David Tilman. 2012. 'Phylogenetic Diversity Promotes Ecosystem Stability'. *Ecology* 93 (8): S223–33. <https://doi.org/10.1890/11-0426.1>.
- Chave, Jérôme, and Egbert G. Leigh. 2002. 'A Spatially Explicit Neutral Model of β -Diversity in Tropical Forests'. *Theoretical Population Biology* 62 (2): 153–68. <https://doi.org/10.1006/tpbi.2002.1597>.
- Clark, James S., David Bell, Chengjin Chu, et al. 2010. 'High-Dimensional Coexistence Based on Individual Variation: A Synthesis of Evidence'. *Ecological Monographs* 80 (4): 569–608. <https://doi.org/10.1890/09-1541.1>.
- Condit, Richard, Peter S. Ashton, Patrick Baker, et al. 2000. 'Spatial Patterns in the Distribution of Tropical Tree Species'. *Science* 288 (5470): 1414–18. <https://doi.org/10.1126/science.288.5470.1414>.
- Dale, Mark R. T. 1999. *Spatial Pattern Analysis in Plant Ecology*. Cambridge Studies in Ecology. Cambridge Univ. Press.
- Dale, Mark R. T., Philip M. Dixon, Marie-Josée Fortin, Pierre Legendre, Donald E. Myers, and Michael S. Rosenberg. 2002. 'Conceptual and Mathematical Relationships Among Methods for Spatial Analysis'. *Ecography* 25 (5): 558–77. <https://doi.org/10.1034/j.1600-0587.2002.250506.x>.
- Eckel, S., F. Fleischer, P. Grabarnik, and V. Schmidt. 2008. 'An Investigation of the Spatial Correlations for Relative Purchasing Power in Baden-Württemberg'. *Asta-Advances in Statistical Analysis* 92 (2): 135–52. <https://doi.org/10.1007/s10182-008-0066-1>.
- Goreaud, François, and Raphaël Pélissier. 1999. 'On Explicit Formulas of Edge-Effect Correction for Ripley's K-function'. *Journal of Vegetation Science* 10 (3): 433–38.
- Goreaud, François, and Raphaël Pélissier. 2003. 'Avoiding Misinterpretation of Biotic Interactions with the Intertype K12 Function: Population Independence Vs Random Labelling Hypotheses'. *Journal of Vegetation Science* 14 (5): 681–92. <https://doi.org/10.1111/j.1654-1103.2003.tb02200.x>.
- Hubbell, Stephen P., Jorge A. Ahumada, Richard Condit, and Robin B. Foster. 2001. 'Local Neighborhood Effects on Long-Term Survival of Individual Trees in a Neotropical Forest'. *Ecological Research* 16: 859–75. <https://doi.org/10.1046/j.1440-1703.2001.00445.x>.
- Jost, Lou. 2006. 'Entropy and Diversity'. *Oikos* 113 (2): 363–75. <https://doi.org/10.1111/j.2006.0030-1299.14714.x>.
- Jurasinski, Gerald, Vroni Retzer, and Carl Beierkuhnlein. 2009. 'Inventory, Differentiation, and Proportional Diversity: A Consistent Terminology for Quantifying Species Diversity'. *Oecologia* 159 (1): 15–26. <https://doi.org/10.1007/s00442-008-1190-z>.
- Losos, Elizabeth C., and Egbert Giles Leigh. 2004. *Tropical Forest Diversity and Dynamism: Findings from a Large-Scale Plot Network*. University of Chicago press.
- Lotwick, H. W., and B. W. Silverman. 1982. 'Methods for Analysing Spatial Processes of Several Types of Points'. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 44 (3): 406–13. <https://www.jstor.org/stable/2345499>.
- Mouquet, Nicolas, Vincent Devictor, Christine N. Meynard, et al. 2012. 'Ecophylogenetics: Advances and Perspectives'. *Biological Reviews* 87 (4): 769–85. <https://doi.org/10.1111/j.1469-185X.2012.00224.x>.
- Pavoine, Sandrine. 2012. 'Clarifying and Developing

- Analyses of Biodiversity: Towards a Generalisation of Current Approaches'. *Methods in Ecology and Evolution* 3 (3): 509–18. <https://doi.org/10.1111/j.2041-210X.2011.00181.x>.
- Pavoine, Sandrine, and Michael B. Bonsall. 2011. 'Measuring Biodiversity to Explain Community Assembly: A Unified Approach'. *Biological Reviews* 86 (4): 792–812. <https://doi.org/10.1111/j.1469-185X.2010.00171.x>.
- Pavoine, Sandrine, Errol Vela, Sophie Gachet, Gérard de Belair, and Michael B. Bonsall. 2011. 'Linking Patterns in Phylogeny, Traits, Abiotic Variables and Space: A Novel Approach to Linking Environmental Filtering and Plant Community Assembly'. *Journal of Ecology* 99 (1): 165–75. <https://doi.org/10.1111/j.1365-2745.2010.01743.x>.
- Pélissier, Raphaël, and François Goreaud. 2015. 'Ads Package for R: A Fast Unbiased Implementation of the K-Function Family for Studying Spatial Point Patterns in Irregular-Shaped Sampling Windows'. *Journal of Statistical Software* 63 (6). <https://doi.org/10.18637/jss.v063.i06>.
- Rajala, T., and J. Illian. 2012. 'A Family of Spatial Biodiversity Measures Based on Graphs'. *Environmental and Ecological Statistics* 19 (4): 545–72. <https://doi.org/10.1007/s10651-012-0200-9>.
- Rao, C. Radhakrishna. 1982. 'Diversity and Dissimilarity Coefficients: A Unified Approach'. *Theoretical Population Biology* 21: 24–43. [https://doi.org/10.1016/0040-5809\(82\)90004-1](https://doi.org/10.1016/0040-5809(82)90004-1).
- Ripley, Brian D. 1976. 'The Second-Order Analysis of Stationary Point Processes'. *Journal of Applied Probability* 13 (2): 255–66. <https://doi.org/10.2307/3212829>.
- Ripley, Brian D. 1977. 'Modelling Spatial Patterns'. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 39 (2): 172–212. <https://doi.org/10.1111/j.2517-6161.1977.tb01615.x>.
- Shen, Guochun, Thorsten Wiegand, Xiangcheng Mi, and Fangliang He. 2013. 'Quantifying Spatial Phylogenetic Structures of Fully Stem-Mapped Plant Communities'. *Methods in Ecology and Evolution* 4 (12): 1132–41. <https://doi.org/10.1111/2041-210X.12119>.
- Shimatani, Kenichiro. 2001a. 'Multivariate Point Processes and Spatial Variation of Species Diversity'. *Forest Ecology and Management* 142 (1-3): 215–29. [https://doi.org/10.1016/S0378-1127\(00\)00352-2](https://doi.org/10.1016/S0378-1127(00)00352-2).
- Shimatani, Kenichiro. 2001b. 'On the Measurement of Species Diversity Incorporating Species Differences'. *Oikos* 93 (1): 135–47. <https://doi.org/10.1034/j.1600-0706.2001.930115.x>.
- Simpson, E. H. 1949. 'Measurement of Diversity'. *Nature* 163 (4148): 688. <https://doi.org/10.1038/163688a0>.
- Stoyan, Dietrich, W. S. Kendall, and Joseph Mecke. 1987. 'Stochastic Geometry and Its Applications'. In *Wiley Series in Probability and Mathematical Statistics*. John Wiley & Sons.
- Whittaker, R. H. 1960. 'Vegetation of the Siskiyou Mountains, Oregon and California'. *Ecological Monographs* 30 (3): 279–338. <https://doi.org/10.2307/1943563>.
- Wiegand, Thorsten, and Kirk A. Moloney. 2004. 'Rings, Circles, and Null-Models for Point Pattern Analysis in Ecology'. *Oikos* 104 (2): 209–29.
- Wiegand, Thorsten, María Uriarte, Nathan J. B. Kraft, Guochun Shen, Xugao Wang, and Fangliang He. 2017. 'Spatially Explicit Metrics of Species Diversity, Functional Diversity, and Phylogenetic Diversity: Insights into Plant Community Assembly Processes'. *Annual Review of Ecology, Evolution, and Systematics* 48 (1): 329–51. <https://doi.org/10.1146/annurev-ecolsys-110316-022936>.